

Partial-swap frames for integer multiplication

Retained triples, controlled blocks, and exact conditional assembly

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With substantial OpenAI GPT-6 Astra and Codex assistance

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Abstract

We record the selected (25, 23, 57) retained-triple construction, its rational partial-swap compiler, compatible carrier histograms, controlled contiguous blocks, and the strengthened complex recurrence. Subject to the pinned upstream analytic and fixed-tape interfaces, the exact parameters give $T(n) = O(n(\log n)^{1-\kappa})$ with $\kappa = 942293/500000000000 = 1.884586 \times 10^{-6}$. The finite arithmetic certificates check counts and inequalities; they are not a formal verification or an implementation of a complete multiplication machine. The common-basis step is a rational generic-existence proof.

1 Scope and attribution

The computational model is a fixed finite-alphabet multitape Turing machine. Every finite network, basis, prime, and tape count is fixed independently of the input length. The bit payload field is $\mathbb{F}_2$; its address geometry is rational and is subsequently reduced at a suitable odd prime. These two fields have different purposes. The complex payload consists of Gaussian dyadics and uses the retained binary address-label interface.

The retained framework is Douglas Colkitt’s compact-control construction [2], with Bortlesboat’s alignment [3], eumemic’s complex producer and Gaussian resampling [4, 6, 7], dleen’s retained totals and stage sharing [5], and Zhihao Chen’s finite-network contribution [8], pinned at 6725c6a17b17871a35353fd29157f4ed851bc114. The upstream multiplication manuscript is pinned at adc7f1241b42e322a6451854ab7e4b4c146bf78a [1]. The partial-swap and selected retained-triple development is by icekylinx, with substantial OpenAI GPT-6 Astra assistance; review and integration used OpenAI Codex assistance. Prior contributions retain their attribution and licenses in NOTICE and SOURCES.json. This submission extends icekylinx’s bulk-recursion PR #10 [9], pinned at 62691e395a0458ce089a1c7b5d89e74291e95e29; its parameterized mixed-width and corrected Gaussian arguments are retained below with the new construction and constants.

2 Partial swaps and the retained scalar producer

2.1 Exact lower-triangular conjugation

For a rational projector P , define

$$D_P = \begin{pmatrix} I - P & P \\ P & I - P \end{pmatrix}.$$

Then $D_P^2 = I$ and $\text{rank}(D_P - I) = \text{rank } P$. For nondegenerate $U \subseteq V$, their orthogonal projectors satisfy $P_U P_V = P_V P_U = P_U$, whence

$$D_{P_V} D_{P_U} = D_{P_V - P_U}.$$

Thus either orientation of a nested frame edge costs its absolute dimension change. A scalar gate whose incident roles share one frame commutes with that frame's common address permutation.

Write the ordered lower-lower elimination as $P = L\Pi R$, where L, R are invertible lower triangular and Π is a partial permutation. Put

$$C = RL, \quad Q_X = \Pi\Pi^t, \quad Q_Y = \Pi^t\Pi, \quad K = Q_Y C - C Q_X, \quad G = \begin{pmatrix} L & 0 \\ 0 & R^{-1} \end{pmatrix} \begin{pmatrix} I & 0 \\ K & I \end{pmatrix}.$$

Idempotence gives $\Pi C \Pi = \Pi$. Multiplication, using $\Pi Q_Y = \Pi = Q_X \Pi$, yields the exact identity

$$G^{-1} D_P G = \begin{pmatrix} I - Q_X & \Pi \\ \Pi^t & I - Q_Y \end{pmatrix}.$$

Both G and G^{-1} are lower triangular in the physical order of the two banks. The displayed matrix is exactly the collection of cross-bank swaps specified by Π . A contiguous identity block of t pivots is therefore one width- t interchange, surrounded by the retained $O(V)$ ordered-affine passes. No reversal or gathering of separated pivots is assumed. Choose the address prime only after fixing all rational bases and factorizations, excluding their finitely many bad denominators and nonzero minors.

2.2 Scalar map and dirty-scratch wrapper

For $h \in \{25, 23, 57\}$ write $v_h = \binom{h}{3}$. There is one input X_T for each triple. In common-point context i put $y_{ab} = X_{\{i,a,b\}}$ and compute

$$A_i = \sum_{a < b, a, b \neq i} y_{ab}, \quad D_{i,\{a,b\}} = \sum_{u < v, u, v \notin \{i,a,b\}} y_{uv}.$$

The $q_h = 3v_h + h$ designated outputs retain all totals. The local cancellation-free producer pairs consecutive vertices, sums coarse edges between pairs, and puts internal edges into the corresponding coarse vertex weights. Recurse on these weighted vertices down to at most two vertices; leave-one and leave-two group sums, together with the surviving strips, partition each desired exclusion sum. This proves correctness by induction. The order in context i lists intact global pairs first, then the unlisted vertices other than i . Intern equal supports; pair-star supports shared across contexts are identified by their actual common mask and union mask. Retain total ancestors and prune unused nodes.

Let V copy physical inputs into source-use roles, L be the reversible addition/fanout circuit, and J inject the designated outputs. For target T , inject the three side outputs indexed by $i \in T$ and the three totals A_i , $i \in T$. If a source triple is S , the first coefficient is $[|S \cap T| = 1]$, and the second is $|S \cap T| \bmod 2$. Their sum in $\mathbb{F}_2$ equals $[S = T]$. Hence $JLV = I$. For arbitrary initial auxiliary vector z , the chronological schedule

$$L, J, L^{-1}, V, L, J, L^{-1}, V$$

injects $JLz + JL(z + Vx) = x$ and restores z . This description treats J, V as elementary copy/injection sequences, which are self-inverse over $\mathbb{F}_2$. Three alternating data shears exchange the two data banks; use the inverse schedule for the middle invocation. Scratch is restored for every initial value, not only zero.

2.3 Positive labels preserving carrier dependencies

Use $H_h = I - J/9$ on $\mathbb{Q}^h$ and triple indicator t_T . Then

$$\langle t_S, t_T \rangle = |S \cap T| - 1, \quad U_i = \{x : \sum_j x_j = 3x_i\}, \quad x^t H_h x = \sum_{j \neq i} x_j^2 \quad (x \in U_i).$$

Thus U_i is positive definite, each triple line has norm 2, and intersection-one triple lines are orthogonal. A side output receives $U_i \cap \{x_a + x_b = 0\}$, of dimension $h - 2$; a total receives U_i , of dimension $h - 1$.

For an original cancellation-free node with common set C and union M , its envelope is

$$E(C, M) = \{x : \text{supp } x \subseteq M, x_i = t \ (i \in C), \sum_j x_j = 3t\}.$$

Identify addition nodes having equal envelopes for label propagation, keeping sources separate. Find the initial carrier matching using these envelopes and add its continuation arcs as label-only dependencies. They respect envelope inclusion and strict event order. Backward intersection of all reachable output constraints gives

$$W_g = \{x : \sum_j x_j = 3x_a \ (a \in F_g), x_b + x_c = 0 \ (\{b, c\} \in E(G_g))\}.$$

Here F_g lists reachable common points and G_g their excluded-pair graph. Sources keep their triple lines. We have $E_g \subseteq W_g$ and $W_g \subseteq W_{g'}$ along every augmented arc. Every retained node reaches an output, so W_g lies in a positive U_a . Nondegeneracy, nesting, retained matches, and side-output orthogonality follow.

For addition nodes $|F_g| \in \{1, 2\}$ and G_g avoids F_g . A nonbipartite component forces zero; each bipartite component, including isolated vertices, supplies a signed parameter z_j . If its color classes are A_j, B_j , the remaining relation is

$$(3 - |F_g|)t = \sum_j (|A_j| - |B_j|)z_j.$$

Consequently $\dim W_g$ is the number of bipartite components on the coordinates outside F_g . Signed component relations also give an exact rational containment test. Orthogonal complements and tensor products of these nondegenerate spaces are nondegenerate; ambient H_h is nondegenerate since $h \neq 9$. Reversed invocations use orthogonal complement frames, so reverse dependencies remain nested.

2.4 Carrier matching and its rank histogram

A donor is an addition u ; a receiver is an individual later operand use or designated output use. A donor can preserve either operand as its nonpivot and continue that operand's carrier to a later use w when $W_u \subseteq W_w$. A matching joins carrier paths without cycles or simultaneous identification of occupied roles. Every value retains an unmatched first use, which supplies its carrier path. Thus

$$R_h = c_h + q_h - |\mathcal{M}_h|.$$

Use events ordered by new dimension, old envelope dimension, and original node/output number. Deterministic Hopcroft–Karp orders donors by node number, first operand before second, and uses by consumer then output order, in both breadth-first and shortest augmenting-path searches. Initial matching arcs were propagated before enlargement; they remain valid with this event order.

For rank r_x and number of uses d_x , the unmatched histogram is

$$H^{(0)}(z) = \sum_{s \text{ source}} d_s z + \sum_{x \text{ addition}} \left((d_x - 1)z^{r_x} + z^{h-r_x} + \sum_{y \text{ operand of } x} z^{r_x-r_y} \right) \\ + \sum_{x \text{ side}} (z^{h-1-r_x} + z) + \sum_{x \text{ total}} (z^{r_x} + z^h).$$

For a match $u \rightarrow w$ preserving y , add

$$-z^{h-r_u} - z^{r_y} - z^{r_w-r_y} + z^{r_w-r_u}.$$

Rank-zero terms are bookkeeping. The selected counts satisfy

h	v_h	c_h	q_h	$ \mathcal{M}_h $	R_h
25	2300	45963	6925	1589	51299
23	1771	34764	5336	1324	38776
57	29260	651213	87837	9264	729786

$$\sum_r r H_{h,r} = h R_h + 2h(h-1).$$

Only retained totals drop, by $h-1$ each. The compatible complete histograms are recorded below (entries start at rank zero).

$$H_{23} = [9840, 38409, 32471, 8898, 9754, 2116, 5737, 1725, 4145, 1547, \\ 3128, 1551, 2893, 1925, 2841, 2450, 2327, 3096, 2344, 3628, 9435, 2110, 23, 23],$$

$$H_{25} = [12884, 50477, 41948, 11341, 13202, 2948, 6911, 1451, 5205, 2429, \\ 3751, 1361, 3720, 2639, 3241, 1571, 3334, 3366, 2963, 4019, 2608, 4868, \\ 12360, 2802, 25, 25],$$

$$H_{57} = [171318, 651573, 566364, 137233, 164158, 15372, 75503, 8071, 52184, \\ 13888, 20431, 7625, 32053, 10364, 19167, 8889, 27939, 11286, 19271, 8785, \\ 21602, 9814, 19261, 8795, 22732, 12389, 19365, 8691, 21012, 10404, \\ 19355, 8701, 22142, 12352, 19459, 8597, 20422, 10196, 19449, 8607, \\ 19956, 16818, 19553, 8503, 19604, 20476, 18289, 8171, 18510, 37106, \\ 16535, 46799, 14592, 52132, 167608, 40250, 57, 57].$$

These histograms count auxiliary transitions within one invocation only. They exclude physical data growth and outer macro transitions.

2.5 Endpoint and asymmetric bank accounting

Use source frame D_{P_U} and destination frame $D_{P_{U^\perp}}$. Their product is D_I , the full address interchange. Together with the scalar bank exchange and arbitrary scratch restoration, this is the retained all-role endpoint identity. In particular the extra source rank charge of additive-shear framing is absent.

Put $N = v_{25}v_{23}v_{57}$ and

$$B_1 = NR_{25}/v_{25}, \quad B_2 = NR_{23}/v_{23}, \quad B_3 = NR_{57}/v_{57}, \quad E = B_3 - B_1, \quad W = 2N + B_2 + B_3.$$

Match all real stage-one roles to a subset of stage-three roles using an intersection-one neighbor permutation of middle-axis triples. The intersection-one graph has a regular bipartite double cover of positive degree; edge counting proves Hall's condition, hence such a permutation exists.

Orthogonality makes each stage-one terminal 25-space a subspace of the matched stage-three entry $(m - 57)$ -space. Thus the paired rank is $m - 57 - 25$; each of the $E \geq 0$ unmatched stage-three roles enters directly with rank $m - 57$. No extra scalar role is introduced. With $m = 25 \cdot 23 \cdot 57 = 32775$,

$$L = N \left(\frac{6}{23} + \frac{6}{21} + \frac{6}{55} \right), \quad s = Wm - 2N + 2L = 190768577823254200, \quad 2N - 2L = 82076640800 > 0.$$

Before the initial inverse-stage copy, raise the physical Y data frame to its common starting frame D_0 and copy with both roles at D_0 . This moves an existing enlargement earlier; all later frames contain D_0 . The direct auxiliary entrance replaces the split entrance with equal rank mass. It gives exactly B_2 copies of class A4 and $2N$ copies of A5 in the following table.

3 One controlled rational basis

3.1 The ordinary projector corner

For a rank- j projector in dimension h with $j > h/2$, put $r = h - j$. In a generic rational basis its first- r /last- r corner is invertible: one witness pairs these coordinates in rank-one blocks $\frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ and is the identity on the middle $h - 2r$ coordinates. Similarity of rational projectors with the same rank proves that the relevant minor is a nonzero rational function of the basis.

Write $I - P = XY$, $YX = I_r$. The corner is $-X_L Y_R$, so X_L, Y_R are invertible. Eliminating it leaves on the middle coordinates

$$I - X_M Y_M - (-X_M Y_R)(-Y_R^{-1} X_L^{-1})(-X_L Y_M) = I_{h-2r}.$$

The resulting lower-lower profile is r singleton pivots and one contiguous block of width $2j - h$. For $j \leq h/2$ retain j singleton pivots; for $j = h$ use one block of width h . The last is still smaller than the full tensor dimension m . Nonzero scalings are absorbed in the triangular factors. This argument is also the large-projector identity used in the controlled corners below.

3.2 Inner boundary prescriptions

Use $F_1 = \mathbb{Q}^{25}$, $F_2 = \mathbb{Q}^{23}$, $F_3 = \mathbb{Q}^{57}$, $H = 575$ and $m = 32775$. For physical inner index $i = 23\alpha + \beta$ set

$$K = T_2(I_{25} \otimes L_2), \quad T_2(u \otimes e_\beta) = J_\beta u \otimes e_\beta, \quad J_\beta = M_\beta L_1,$$

where L_1, L_2 are free invertible rational bases. Put $n = 82$, $I = (0, \dots, 24)$, $\Sigma = (0, \dots, 6)$ and

$$R = [I, \Sigma, I, I], \quad C = [I, I, \Sigma, I].$$

For $i < n$ prescribe $e_\alpha^t M_\beta = e_{R_i}^t$; for $i \geq H - n$ prescribe $M_\beta^{-1} e_\alpha = e_{C_{i-(H-n)}}$. The latter is equivalently a fixed column of M_β . The effective starting residues modulo 23 are respectively $\{0, 2, 9, 11\}$ and $\{10, 12, 14, 21\}$. They are pairwise distinct and disjoint. No coefficient coordinate is prescribed twice on one side, nor shared across the two sides. The prescribed row positions are near the start and the prescribed column positions near the end. They are disjoint, so every M_β completes to a permutation matrix. Their unspecified entries form an affine space; its invertible locus is nonempty and irreducible.

The first 25 rows and last 25 inverse columns give matching L_1 and L_1^{-1} bases, so every stage-one single-active-factor projector has an ordinary 25-corner. For stage two, the first factor is rank one; its row/column weights separate, leaving a diagonal rescaling of $L_2 B L_2^{-1}$ in the first-23/last-23 corner. Thus all ordinary 23-dimensional profiles are available in the same family.

3.3 The A5 tree certificate

The inner projector $(I - P_1) \otimes (I - P_2)$ has rank 528 and nullity 47. Eliminate the first 23 rows of its nullspace restriction. The remaining 24 weighted incidence vectors on $\{0, \dots, 24\}$ have forward edges

$$(0, 23), (1, 24); \quad ((j+2) \bmod 23, j) \ (0 \leq j < 7); \quad ((j+9) \bmod 23, j) \ (0 \leq j \leq 14)$$

and dual edges

$$(23, 0), (24, 1); \quad (2 + ((-9+j) \bmod 23), j) \ (0 \leq j < 7); \quad (2 + ((-11+j) \bmod 23), j) \ (10 \leq j \leq 24).$$

Both are trees. Here are leaf-removal orders, each leaving vertex 24:

$$\begin{aligned} \text{forward: } & 15, 16, 7, 17, 8, 6, 18, 9, 19, 10, 20, 11, 21, 12, 22, 13, \\ & 4, 2, 23, 0, 14, 5, 3, 1; \\ \text{dual: } & 7, 8, 9, 16, 0, 17, 1, 18, 2, 11, 20, 4, 13, 22, 6, 15, \\ & 23, 14, 5, 21, 12, 3, 19, 10. \end{aligned}$$

For nonzero line coordinates, diagonal rescaling converts these into ordinary tree-incidence vectors. Leaf elimination proves rank 24. Both 47-dimensional nullspace restrictions are therefore nonsingular; the large-projector identity gives 47 singleton pivots and one contiguous block $575 - 94 = 481$.

3.4 Matched outer basis and the paired A1 corner

Choose

$$S = T(I_{57} \otimes K), \quad T(u \otimes e_i) = G_i u \otimes e_i,$$

with a free $L_0 \in \text{GL}_{57}(\mathbb{Q})$. Prescribe the first 57 outer row functionals in the L_0 coordinate order and the next 25 as repetitions of its first 25 coordinates. Prescribe the last 57 inverse columns in matching L_0^{-1} order and the preceding 25 as repetitions of its first 25 coordinates. The first and last 82 multiplicity intervals are disjoint and act on disjoint G_i , giving nonempty rational completion families. Other entries remain free. In particular the ordinary 57-dimensional stage-three corners and the full 57-block for an unmatched entrance are retained.

For a paired role the null projector is

$$Q = I_{57} \otimes P_q + P_t \otimes P_U, \quad \text{rank } P_q = 1, \quad \text{rank } P_U = 25, \quad P_q P_U = P_U P_q = 0.$$

It has rank 82 and the actual A1 projector is $I - Q$. Factor $Q = XY$, $YX = I_{82}$, and consider the first-82/last-82 corner. In the first 25 rows against the rightmost 25 columns, the outer coordinate groups are disjoint, so $I_{57} \otimes P_q$ contributes zero. The $P_t \otimes P_U$ contribution is a generically nonzero diagonal. Normalize and eliminate this contiguous 25-block. These pivots exhaust its 25-dimensional coefficient contribution. The next seven rows have a diagonal seven-block and entries only to its left, supplying the next contiguous block. The residual 50-square corner is a two-by-two matrix of diagonal 25-blocks: its top-right diagonal is generically nonzero, and its determinant is nonzero, so it supplies two more contiguous 25-blocks.

Here is the nonvanishing argument in the prescribed parameter family. The duplicated outer groups in X compare primal coefficient ratios at multiplicity shift $57 \bmod 23 = 11$; those in Y compare dual ratios at shift $25 \bmod 23 = 2$. Distinct coefficient positions are free and the ratios are generically distinct. Orthogonality of the two motif lines imposes their bilinear pairing, not

equality of these distinct coordinate ratios. Thus both restrictions of X, Y are invertible. After the first block elimination, the second diagonal pivot has a reciprocal-first-pivot term with nonzero coefficient in the free outer rank-one coordinates, so it is not identically zero. The last pivot then follows from full nonsingularity. The large-projector identity adds the untouched middle block of width $m - 164 = 32611$. The complete ordered A1 profile is consequently

$$25, 7, 25, 25, 32611, \quad 25 + 7 + 25 + 25 + 32611 = m - 82.$$

3.5 Remaining macro corners and simultaneous choice

A2 is $(I - P_t) \otimes I_H$: its first- H /last- H corner is diagonal, giving 575, 31625. A4 has an identity second inner factor, so its first-23/last-23 corner is diagonal with generically nonzero entries, giving 23, 529.

For A3 the top-right 57-square corner within the H -coordinate block is a diagonal rescaling of an ordinary rank-56 projector. Its first and last 57 multiplicity intervals are disjoint; the diagonal tensor contribution therefore vanishes there. It supplies a singleton and a 55-block. After these exhaust the rank-56 perturbation, the untouched matching interval supplies $H - 114 + 2 = 463$ diagonal pivots as one block. The remaining finite nullity corner supplies 112 singleton pivots; the large-projector middle block has width $m - 2H - 114 + 2 = 31513$. This gives 113 singleton pivots and blocks 55, 463, 31513.

Every additional required minor is a nonzero polynomial after clearing determinants from denominators: ordinary corners have the explicit projector witnesses above; diagonal corners use independent nonzero line coordinates; A5 uses the two trees; A1 uses the ratio and pivot argument. There are only finitely many producer edges. Their product with all invertibility determinants is nonzero on the same irreducible rational parameter family. A rational point avoiding it exists and can be found by enumerating rational parameters and testing these minors. This is fixed-machine preparation independent of recursive widths.

3.6 Complete child list and exact bit moment

With B_i, E, N as above, the macro classes are

class	multiplicity	child widths
A1	B_1	25, 7, 25, 25, 32611
unpaired	E	57, 32661
A2	B_2	575, 31625
A4	B_2	23, 529
A3	$2N$	113 singletons; 55, 463, 31513
A5	$2N$	47 singletons; 481

Physical data growth adds $2N$ copies of each pair $(1, 23)$, $(1, 21)$, $(1, 55)$. For each $h \in \{25, 23, 57\}$ and each histogram rank j , replicate its ordinary profile $NH_{h,j}/v_h$ times. If $j > h/2$ it is $h - j$ singletons and one width- $(2j - h)$ block; otherwise it is j singletons. Width-one blocks are folded into $c_1 = 179603173231020$. These prescriptions define every c_t exactly, without a separate choice of producer histogram. Their rank sum is

$$\sum_t c_t t = s = Wm - 2N + 2L.$$

All widths are strictly less than m . Define

$$\Psi(x) = \frac{\sum_t c_t t^x}{Wm^x}, \quad a_b = \frac{188459}{50000000000}, \quad \tau = 1 - a_b.$$

For completeness the exact rational enclosure used for the certificate is as follows. For $1 \leq y \leq 2$, put $z = (y - 1)/(y + 1)$ and

$$L(y) = 2 \sum_{j=0}^{23} \frac{z^{2j+1}}{2j+1} + \frac{2z^{49}}{49(1-z^2)} \geq \log y.$$

Extract powers of two in m/t and round the resulting upper logarithm up to a multiple of 10^{-12} , calling it ℓ_t . Since $e^u \leq 1 + u + u^2/(2(1 - u/3))$ for $0 \leq u < 3$,

$$\Psi(\tau) \leq \sum_t \frac{c_t t}{Wm} \left(1 + a_b \ell_t + \frac{(a_b \ell_t)^2}{2(1 - a_b \ell_t/3)} \right) < 1 - 6.1778 \times 10^{-14}.$$

The first inequality follows termwise from $(t/m)^{1-a_b} = (t/m) \exp(a_b \log(m/t))$; the final comparison is an exact rational certificate. Together with the integer-width and row construction below, it supplies the retained interchange interface with exponent τ .

4 Integer widths and complete bit rows

The following retained generic mixed-width proof is used with the new partial-swap child list. Its ordered-affine passes are the triangular wrappers proved above; a contiguous partial permutation is implemented directly by an interchange. None of the older numerical moment values is used.

4.1 Integer widths, bounded row padding, and mixed contractions

The batched residual construction uses a fixed finite list of child sizes, not only powers of one radix. The following implementation supplies the corresponding integer-width recursion. All constants in this subsection are fixed finite circuit constants, independent of the input length.

A general moment condition. Let $m, W \geq 2$ be fixed. At one circuit invocation, let the list $t_1, \dots, t_J$ contain integers in $[1, m - 1]$. A child of type i has width $t_i f$, where $f = \lfloor e/m \rfloor$, and operates on exactly $1/W$ of the parent's logical volume. Repetitions in the list represent separate calls. An ordinary rank-one call has $t_i = 1$; a batched block has the corresponding larger value of t_i . Full-rank edges may retain their original m rank-one calls, so no same-width recursive call is needed. Put

$$\Psi(x) = \frac{1}{W} \sum_{i=1}^J (t_i/m)^x, \quad t_* = \max_i t_i < m.$$

If $\Psi(\tau) < 1$ for some $0 < \tau < 1$, then the normalized recurrence

$$F(e) \leq \frac{1}{W} \sum_{i=1}^J F(t_i \lfloor e/m \rfloor) + C, \quad F(e) = O(1) \quad (1 \leq e < m),$$

has $F(e) = O(e^\tau)$ for every positive integer e . Indeed, substituting $F(u) \leq Au^\tau$ into an internal call gives at most $A\Psi(\tau)e^\tau + C$. Choose A to cover the finite base range and $C/(1 - \Psi(\tau))$. Floors can only decrease the right-hand side.

A finite integer depth bound. Define

$$D(e) = 0 \quad (e < m), \quad D(e) = 1 + D(t_* \lfloor e/m \rfloor) \quad (e \geq m).$$

This is an integer algorithm: repeatedly replace e by $t_* \lfloor e/m \rfloor$ until it is below m . The function D is nondecreasing, every child has depth bound at most $D(e) - 1$, and $D(e) = O(\log(2e))$, because each internal replacement is at most $(t_*/m)e$. A row count divisible by $W^{D(e)}$ is therefore sufficient for every root-to-leaf path. At each invocation write the row index as $Wg + w$ and split by w . Every child has complete within-row ranges, logical volume exactly V/W , and a remaining row count divisible by $W^{D(e)-1}$. No padding occurs inside this recursion.

Bit interchange with a remainder. At an integer-width call write $e = mf + r$, $0 \leq r < m$. If $f = 0$, interchange the bounded number of digits individually. Otherwise separate the high r digits of each of the two target chunks. With G denoting all intervening spectators, the layout is

$$(H^+, H^-, G, D^+, D^-).$$

First exchange corresponding digits of H^+, D^+ and move H^+ just before H^- . The layout becomes (D^+, H^+, H^-, G, D^-) . Apply the finite circuit to the two length- mf chunks H^-, D^- . Finally move H^+ back immediately before the final H^- , obtaining

$$(D^+, D^-, G, H^+, H^-).$$

All other fields are preserved. These moves use fewer than a fixed multiple of m single-digit operations and cost $O(V)$. The main chunks split into m equal width- f fields, so every rational shear and every batched contiguous-block shear has exactly the required field shape. A child retains the unused main digits and all remainders as spectators. Thus the preceding moment recurrence applies literally at every integer width; there is no per-level numerical padding of either chunk range.

Supplying the bit rows from existing digits. For an arbitrary width e , compute $D(e)$ and choose the least nonnegative integer ρ such that

$$q^{2\rho} \geq W^{D(e)}.$$

Both numbers can be constructed by fixed-radix integer arithmetic. Since $D(e) = O(\log e)$, we have $\rho = O(\log e)$. For sufficiently large e , $2\rho < e$. The exceptional set of smaller widths is fixed and finite; individual-digit interchange there is absorbed by the asymptotic constant.

Exchange the high ρ digits of the two chunks, then gather their two high parts into one row field before both remaining chunks, exactly as in the preceding layout operation. This row field has range $q^{2\rho}$. Within each original prefix, pad its range to

$$R' = W^{D(e)} \left\lceil q^{2\rho} / W^{D(e)} \right\rceil < 2q^{2\rho}.$$

Every added row includes the entire remaining address set. Recurse once at the remaining width $e - \rho$, which has depth bound at most $D(e)$; then remove the added rows and undo the gathering move. Each completed invocation preserves its row index, so padded zero rows are zero again when they are removed.

The possibly large exponent in $W^{D(e)} = e^{O(1)}$ does not multiply the logical volume by that polynomial. The row digits were already part of the two input chunks. Only rounding the row count

adds data, by a factor strictly below two, once at the outer boundary. Exchanging and gathering $O(\log e)$ high digits costs $O(V \log(2e))$, which is $O(Ve^\tau)$. The radix conversion from binary chunk ranges to the fixed radix q retains the original constant factor, less than q^2 . Consequently the batched construction is an arbitrary-width interchange procedure with the asserted exponent τ and no additional power of its volume or width.

Fixed tapes and descriptors. Use the original depth-first scheduler: park only the other role streams at the active stack top, run the child on one complete role stream, and restore the parked streams on return. The number of child types is fixed, so parking, copying, cleanup and the finite arithmetic transforms contribute $O(V)$ per node. Before descending, retain the two current chunks and merge consecutive spectator fields; the descriptor has a fixed number of field intervals. Its integer lengths have $O(\log V_0)$ bits. All descendants retain the complete ranges of both root main chunks, even when most digits have become spectators. Their volume is therefore at least $q^{2(e-\rho)}$ in the outer construction, while $j \log W = O(\log e)$ at depth j . Hence $\log V_0 = \log V_j + j \log W = O(\log(2V_j))$. Polynomial descriptor work is absorbed by $O(V_j)$, as in the original fixed-tape proof. The larger fixed constant in the depth bound creates no new tape and no growing alphabet.

5 Complex residuals and the precision guard

5.1 Five macro classes and the internal histogram

Keep the retained complex producer with

$$h = 28, \quad v = 3276, \quad m = 21952, \quad N = v^3 = 35158608576, \quad R_c = 93838,$$

$$B = v^2(R_c + 29) = 1007397164592, \quad W = 2N + 2B = 2085111546336.$$

The 29 center roles are included. The whole-residual macro classes are

t	21896	21168	21141	756	729
n_t	B	B	$2N$	$B - N$	$3N$

and before internal batching the remaining singleton count is 91318626008064. The per-invocation internal histogram, indexed from zero, is

$$H_c = [13776, 93912, 23891, 21224, 31667, 13552, 9128, 9240, 10912, 5656, \\ 8456, 4284, 7598, 4872, 7448, 4536, 7803, 4228, 8232, 4872, 7482, 7224, \\ 5901, 8400, 11550, 1512, 5180, 0, 87].$$

It satisfies $\sum_r r H_c[r] = 2629900 = 28(R_c + 29) + 2 \cdot 28 \cdot 29$. For each $r \geq 2$, replace $3v^2 r H_c[r]$ singleton rank units by $3v^2 H_c[r]$ width- r children. All five macro counts remain unchanged. The resulting singleton count is $n_1 = 9668617358400$, and

$$\sum_t n_t t = 45772350635112192.$$

The retained residual basis map puts all r directions into a contiguous prefix. Their rf selected axes are one child. If directional signs differ, the exact identity $C^{-1} = -iZCZ$ gives

$$\bigotimes_{j=1}^r (C^{\varepsilon_j})^{\otimes f} = (-i)^{f|B_-|} Z_{B_-} C^{\otimes rf} Z_{B_-}.$$

The two address-sign passes and fourth-root phase cost $O(V)$, make no recursive call, and contribute only a fixed number of coefficient operations per value. This also covers the uniform-sign internal residuals; it does not assume that arbitrary binary residual directions always have identical signs.

Set $a_c = 417/100000000$, $\sigma = 1 - a_c$. Using the same rational logarithm upper bounds $\ell_t \geq \log(m/t)$ as for the bit moment,

$$\mathcal{C}(\sigma) = \frac{1}{W} \sum_t n_t (t/m)^\sigma \leq \sum_t \frac{n_t t}{Wm} \frac{1}{1 - a_c \ell_t} < 1 - 2.3982 \times 10^{-11}.$$

All denominators are positive. The bound $e^u \leq (1 - u)^{-1}$ for $0 \leq u < 1$ makes the final comparison entirely rational.

5.2 A uniform dependency-path guard

The retained scalar path argument counts both orientations of frame edges and paths changing wires at a gate: the total residual rank on one arithmetic dependency path is at most $q = m + 6h = 22120$. Take $\rho = 6/5$. The exact integer inequality $160000^5 < m^6$ gives $m^\rho > 160000$, and at most one residual can exceed $q/2$. If it is 21896, the remaining rank is at most 224; no 729- or 756-residual fits, so all remaining ranks are at most 28. Hence

$$\sum_{\text{path}} (r/m)^\rho < (391/392)^{6/5} + 448/160000 < 0.996941 + 0.0028 < \gamma := 4999/5000.$$

The middle bound can be checked rationally from $(1 - u)^{6/5} \leq 1 - 6u/5 + 3u^2/(25(1 - u))$, at $u = 1/392$. If the large residual is 21168 or 21141, the remaining rank is at most 979 and remaining widths at most 756; since $756^{1/5} < 4$,

$$\sum_{\text{path}} (r/m)^\rho < 27/28 + 4 \cdot 979/160000 < 0.989.$$

If none of those three large residuals occurs, every residual is at most 756 and the bound is $4q/160000 < 0.554$. Thus γ is uniform for the five macro classes and all internal children together.

Stop at width $e < d^\beta$, $0 < \beta < 1$, and use elementary kernels there. At an internal node write $e = mf + r$, $r < m$. Process the r remainder axes individually, adding their bounded work to the local constant, and recurse on widths tf . For large d , every stopping leaf descended from an internal node has width at least $d^\beta/(2m)$. The local dependency allowance, including the sign wrappers, satisfies

$$36W^3 + 4s + 4W + 8m + 4 < E := 64(W + m + 1)^3.$$

Put

$$C_{\text{dep}} = 5000(E + 16m + 1).$$

Strong induction using the path guard gives

$$A(e) \leq C_{\text{dep}} d^{-\beta(\rho-1)} e^\rho.$$

Indeed a leaf has $A(e) \leq 8e \leq 16m d^{-\beta(\rho-1)} e^\rho$; the child sum uses at most γ of the asserted internal bound, and its remaining gap covers E since $e \geq d^\beta$. A root already below threshold is handled directly. For rational $\zeta > 0$ put

$$C_1 = \rho - (\rho - 1)\beta + \zeta, \quad C_0 = \lceil 128m(1 + 1/\zeta)C_{\text{dep}} \rceil, \quad \Delta = \lceil C_0 d^{C_1} \rceil.$$

The retained elementary-depth induction, including reserved axes, outer phases and conversion, then gives $p + \Delta$ fractional bits and $\Delta + 3$ integer/sign bits. If $\epsilon C_1 < 1$, their width is $O(p)$.

5.3 Rows, stopping, and transfer to the layer interface

Let $t_* < m$ be the largest child and compute $D(e) = 0$ for $e < m$ and $D(e) = 1 + D(t_* \lfloor e/m \rfloor)$ otherwise. It is $O(\log e)$ and bounds every descendant depth. With the retained compact-control parameters, reserve

$$q_0 = \lceil \log_2 W \rceil D(d), \quad Q_0 = q_0 + q_F + q_B.$$

Process the first $q_0 + q_F$ and last q_B selected axes individually; if at most Q_0 axes exist, process them all individually. Otherwise the middle active interval is consecutive. Only the q_0 row chunks supply row indices, separate from the two compact fields. Pad their count once to a multiple of $W^{D(d)}$, by a factor below two. Every child retains complete compact and spectator ranges and has exactly $1/W$ of its parent's volume. No internal padding is needed. The reserved-axis bound remains

$$Q_0 = O(\log(2d) + d^{\max\{1-c, 0\}} \log p + 1).$$

Remainders commute with the completed tensor map and cost $O(V)$ per node. The fixed-tape depth-first scheduler and complete-row argument of the bit construction apply with the retained compact descriptors.

Since $\mathcal{C}(\sigma) < 1$, the total leaf σ -weighted width, including the volume factor W^{-j} at depth j , is at most d^σ . Thus leaf work is $O(d^{\sigma+\beta(1-\sigma)})$. Internal compact operations cost $O(G^\tau e^\tau + 1)$ per current volume. Summing the strict moments gives the retained complete normalized layer bound

$$O\left((\log p)^3 \left[d^\chi + d^{\sigma+\beta(1-\sigma)} + d^{\max\{1-c, 0\}} + 1\right]\right), \quad \chi = \tau + (1 - \beta) \max\{\sigma - \tau, 0\}.$$

For $\sigma \leq \tau$, use $\mathcal{C}(\tau) \leq \mathcal{C}(\sigma)$; otherwise use $e^\tau \leq d^{\beta(\tau-\sigma)} e^\sigma$ at each internal node. These arguments do not assume equal leaf depth. The usual strict conditions on λ, λ' therefore give the retained $O(Vd^{\lambda'})$ layer interface, now with the new moment and the new C_{dep} .

6 Exact conditional assembly

Theorem 6.1 (Conditional fixed-tape multiplication bound). *Assume the pinned upstream multiplication theorem and retained analytic, ordered-affine, compact-control, Gaussian-map, and fixed-tape interfaces specified in the opening section. The partial-swap interchange interface and the complex layer proved here may be substituted into those interfaces. They give*

$$T(n) = O\left(n(\log n)^{1-\kappa}\right), \quad \boxed{\kappa = \frac{942293}{500000000000} = 1.884586 \times 10^{-6}}.$$

Proof. Take the following exact parameters:

$$\begin{aligned} a_b &= \frac{188459}{50000000000}, & a_c &= \frac{417}{100000000}, \\ \tau &= 1 - a_b, & \sigma &= 1 - a_c, \\ \epsilon &= \frac{1}{2 + a_b} = \frac{50000000000}{100000188459}, & c &= 1, \\ \beta &= \frac{1}{1000}, & \delta &= \frac{1}{10^{16}}, \\ \lambda &= \tau + \frac{1}{10^{20}}, & \lambda' &= \tau + \frac{2}{10^{20}}, \\ \zeta &= \frac{1}{10000}, & C_1 &= \frac{11999}{10000}. \end{aligned}$$

The strict bit and complex moments were certified above. Since $\sigma < \tau$, the internal layer exponent is $\chi = \tau$. The complex leaf saving is

$$(1 - \beta)a_c = \frac{416583}{10^{11}} = 4.16583 \times 10^{-6} > a_b.$$

Thus both the internal and leaf exponents are strictly below λ' , and the displayed gaps between τ, λ, λ' absorb fixed logarithmic factors. The precision guard has $C_1 = 6/5 - \beta/5 + \zeta$ and $1 - \epsilon C_1 > 0$. Compact control allows $c = 1$; its reservation exponent is zero and $1 - \epsilon(1 + c) = 1 - 2\epsilon > 0$ gives $K = o(\ell)$ with $K = d$.

For the retained Gaussian parameters $\alpha = \lfloor \sqrt{b/(8d)} \rfloor$ and $\gamma_G = 2d\alpha^2 \leq b/4$, the same positive $1 - 2\epsilon$ ensures the eventual conditions $b \geq 64d^2$, $\alpha^2\theta_i \geq 1$, and prime-interval growth. Use the corrected chirp input bound $F = 2^{\lceil 1.14\alpha^2 \rceil}$; the following section proves that $P = 34p$ still suffices. The retained target errors and final rounding therefore apply.

The seven assembly savings are the exact rational numbers

$$\begin{aligned} g_1 &= 1 - 2\epsilon, & g_2 &= \epsilon a_b, & g_3 &= \epsilon(a_b - 2/10^{20}), \\ g_4 &= (1 - \epsilon)a_b, & g_5 &= 1 - \delta - 2\epsilon, & g_6 &= 1 - \delta - \epsilon, & g_7 &= \epsilon. \end{aligned}$$

Writing $b_0 = a_b/(2 + a_b)$ gives $g_1 = g_2 = b_0$, $g_3 = b_0 - 2\epsilon/10^{20}$, $g_4 > b_0$, and $g_5 = b_0 - \delta$. Since $\delta > 2\epsilon/10^{20}$, g_5 is the least of the seven. Exact rational subtraction gives

$$\min_i g_i - \kappa = \frac{a_b}{2 + a_b} - \frac{1}{10^{16}} - \frac{942293}{500000000000} > \frac{44822}{10^{17}} > 0.$$

All retained parameter conditions are strict with this choice; in particular the finite arithmetic certificate evaluates the complete 29-condition interface and these seven margins with rational arithmetic. The final positive gap absorbs remaining logarithmic factors, proving the claimed bound in the retained model. $\square$

The count and moment data are finite arithmetic certificates for the specified producer and common-basis construction. The theorem's use of the pinned analytic interfaces and its fixed rational basis preparation is separate from checking those integers and rational inequalities.

7 Gaussian input-error correction

The retained fast Gaussian construction generates $2^{-B}e^{\pi\alpha^2\beta_j^2}$ at precision P , where $B = \lceil 1.14\alpha^2 \rceil$, then shifts by B and multiplies by the disk-grid input u_j . The exponential interface gives absolute error at most 2^{1-P} before the shift. Thus the error after shifting is controlled by 2^{B+1-P} , not by a constant multiple of $2^{-P}e^{\pi\alpha^2/4}$ uniformly in α .

Set $F = 2^B$ in the retained chirped-correlation block lemma. Since $|\beta_j| \leq 1/2$ and $\pi/(4\log 2) < 1.14$, this bounds the exact input magnitude. Including the product and rounding, it also gives the required input accuracy $|\tilde{f}_j - f_j| \leq 2^{4-P}F$. The distinction matters: $2^B/e^{\pi\alpha^2/4}$ is unbounded, so the smaller exact-magnitude bound alone does not justify this accuracy.

No additional asymptotic precision is needed. For the inherited integer parameters $p > 100$ and $2 \leq \alpha < \sqrt{p}$, let $L_A = 3\lceil \sqrt{p}/(2\alpha) \rceil + 1$ and $\Gamma = 29p$. Then $L_A \leq p$, $\lceil \log_2 L_A \rceil \leq p - 1$ and $B \leq 1.14p + 1$. The block lemma requires

$$P \geq p + \Gamma + \lceil \log_2(FL_A) \rceil + 11.$$

Its right side is at most $32.14p + 11 < 34p$, so the retained $P = 34p$ satisfies the corrected error budget. The packing widths remain $O(p)$, and the target errors, tape-time exponents and final rounding guarantee are unchanged. The combined manuscript patch includes this correction directly in the retained Gaussian proof.

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